

Practical 5

Title – Coupled Analysis – Static + Thermal

Sample Example

Ex.3.10.1 : An axial load of 400 kN is applied at 20°C to the rod, as shown in Fig. P. 3.10.1(a). The temperature is then raised to 50°C. The material for element 1 is aluminium and for element 2 is steel. The relative properties for element 1 and 2 are as shown below.

Property	Element 1	Element 2
Material	Aluminium	Steel
Modulus of Elasticity	$70 \times 10^9 \text{ N/m}^2$	$200 \times 10^9 \text{ N/m}^2$
Cross-section Area	900 mm^2	1200 mm^2
Coefficient of Thermal Expansion	$2.3 \times 10^{-6} \text{ per } ^\circ\text{C}$	$11.7 \times 10^{-6} \text{ per } ^\circ\text{C}$

Determine :

- (i) the element stiffness matrix;
- (ii) the load vector;
- (iii) the nodal displacements; and
- (iv) the stresses in each element.

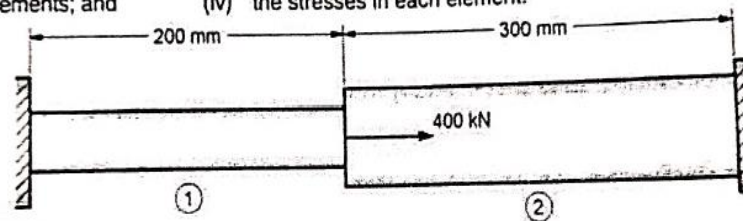


Fig. P. 3.10.1(a)

Soln. :

Given : $A_1 = 900 \text{ mm}^2$; $A_2 = 1200 \text{ mm}^2$; $l_1 = 200 \text{ mm}$;
 $l_2 = 300 \text{ mm}$; $E_1 = 70 \times 10^3 \text{ N/mm}^2$; $E_2 = 200 \times 10^3 \text{ N/mm}^2$;
 $\alpha_1 = 2.3 \times 10^{-6} \text{ per } ^\circ\text{C}$; $\alpha_2 = 11.7 \times 10^{-6} \text{ per } ^\circ\text{C}$; $\Delta T = 50 - 20 = 30^\circ\text{C}$;
 $P_2 = 400 \times 10^3 \text{ N}$.

Procedure

Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

Step 2 Pre-processor

Pre-processor – Element Type – Add – Link – 3D finite stn – 180 - OK –

Close

Step 3 Material

Material Properties – Material Models – Structural – Linear – Elastic – Isotropic – Ex – $7e4$ – PRXY – 0.3 – Thermal Expansion – Secant Coefficient – Isotropic - Conductivity – Isotropic – ALPX – $2.3e-6$ – OK

Material – New model – ID 2 – OK - Structural – Linear – Elastic – Isotropic – Ex – $2e5$ – PRXY – 0.3 – Thermal Expansion – Secant Coefficient – Isotropic – ALPX – $11.7e-6$ – OK

Step 4 Section

Sections – Link – Add – 1D – 1 – Name – 1 – Area – 900 – Apply – 1D – 2 - Name – 2 – Area – 1200 - OK

Step 5 Modeling

Modeling – Create – Key point – In Active CS – 1 - 0,0,0 – Apply – 2 – 200,0,0 - Apply – 3 – 500,0,0

Modeling – Create – Lines – Straight Lines – Select Key points 1 & 2, 2 & 3

Step 6 Meshing

Meshing – Mesh Attributes – Picked Lines – Select Line 1 – Ok – Select Material 1 – select element number 1 LINK 180 – select section 1 – apply – select line 2 – OK - Select Material 2 – select element number 1 LINK 180 - select section 2 – OK

Size Control – Manual Size – Lines – Picked Lines – Pick All – NDIV – 1 – OK

Mesh – Lines – Pick All

Step 7 Loads

Loads – Define Loads –Apply – Structural – Displacement – On Key points

– Select Key point 1 & 3 – OK – All DOF – 0 – OK

Loads – Define Loads –Apply – Structural – Force – On Key points – Select

Key point 2 – OK – FX – $4e5$ – OK

Loads – Define Loads –Apply – Temperature – on lines – select both lines –

OK – Temperature – 30 - OK

Step 8 Solution

Solution – Solve –Current LS - OK

Step 8 General Results

General Post Process – List Results - Reaction Solution – All Items – OK

Nodal Solution – DOF Solution – Displacement Vector Sum

Element Table – Define Table – ADD – By Sequence Num – SMISC,1 –

Apply – ADD – By Sequence Number – LS,1 – OK

List Results – Element Table Data – Select SMISC1 and LS1 - OK